Homework 8: Fibers & Chromatography

Paper track. Pairs with Home Lab 8.

Part A: Burn-test flow chart (2 pts each)

Name the class (animal / vegetable / synthetic) and a likely specific fiber:

1. Shrinks away from the flame before it touches, melts and drips, hard shiny bead, sharp chemical smell.
2. Burns steadily with an afterglow, smells like burning paper, soft gray ash, no bead.
3. Curls away, sputters, self-extinguishes, smells like singed hair, brittle black bead.
4. Smooth and shiny (not scaly), burns slowly, self-extinguishes, smells like burning hair.
5. Burns very fast with almost no melting, smells like paper, very little ash.

Part B: Fiber facts (2 pts each)

1. Where do most synthetic fibers come from? What about rayon and acetate?
2. A running shirt is advertised to "wick moisture." What fiber class is it, and how does the wicking actually work?
3. A carpet fiber goes from the carpet to the victim to a suspect who never touched the carpet. What is this called?

Part C: Rf calculations (2 pts each)

1. A dye spot moved 2.4 cm; the solvent front moved 8.0 cm. Rf?
2. A dye spot moved 7.2 cm; the solvent front moved 8.0 cm. Rf? Is this dye more or less soluble in the solvent than the dye in #1?
3. A spot stayed exactly on the start line. Rf?
4. Can an Rf ever be greater than 1? Why or why not?

Part D: Matching a pen (4 pts)

The crime-scene note's ink separates into **3 spots** with Rf values **0.30, 0.55, 0.80**. Three suspects' pens give:

- Pen A: 2 spots at Rf 0.30, 0.80
- Pen B: 3 spots at Rf 0.30, 0.55, 0.80
- Pen C: 3 spots at Rf 0.25, 0.50, 0.75

Which pen wrote the note? Why not the others?